

Feistel Cipher & DES Overview

1. The Feistel Cipher Structure

The Feistel cipher is a symmetric structure used in the design of many block ciphers. It processes a plaintext block by splitting it into two halves and applying a series of rounds.

Encryption Formulas (Round i)

$$\begin{aligned}L_i &= R_{i-1} \\ R_i &= L_{i-1} \oplus f(R_{i-1}, K_i)\end{aligned}$$

Decryption Formulas

Decryption uses the same structure but applies subkeys in reverse order (K_n to K_1):

$$\begin{aligned}R_{i-1} &= L_i \\ L_{i-1} &= R_i \oplus f(L_i, K_i)\end{aligned}$$

2. Data Encryption Standard (DES)

DES is a 16-round Feistel cipher that uses a 56-bit key to encrypt 64-bit blocks of data.

Component	Specification
Block Size	64 bits
Key Size	56 bits (effective)
Rounds	16 Rounds
Round Function	Expansion, Key Mixing, S-box Substitution, P-box Permutation

3. Role of the S-Box

The S-box (Substitution-box) is the only **non-linear** component in DES, providing the primary cryptographic strength (Confusion).

S-Box Operation

- **Input:** 6 bits $\{b_1, b_2, b_3, b_4, b_5, b_6\}$
- **Row Selection:** Bits $\{b_1, b_6\}$ (Outer bits)
- **Column Selection:** Bits $\{b_2, b_3, b_4, b_5\}$ (Inner bits)
- **Output:** 4-bit value from the substitution table.

Example S-Box Concept:

By mapping 6 bits to 4 bits non-linearly, the S-box ensures that small changes in the key or plaintext result in significant, unpredictable changes in the ciphertext.